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Geologic and paleomagnetic evidence for a possible Miocene collision in western Panay, central Philippines

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ABSTRACT

The Palawan metamorphic terrane, represented by Paleozoic basement on the northwestern tip of Panay, central Philippines, collided with the remainder of the island between late Oligocene and middle Miocene time. Subduction continued north and south of the collision zone, along the Manila and Negros trenches, respectively. Evidence for the collision is (1) the juxtaposition along a major fault of old melange terrane against highly metamorphosed Paleozoic continental sediments similar to those exposed on Palawan island itself, and (2) Miocene thrusting in the melange terrane oriented parallel to the suture.

Paleomagnetic studies show that since the early Miocene the island of Panay has rotated 20° clockwise (declination 20.9°, inclination 22.1° for six sites), whereas the northern Philippines has rotated counterclockwise. These contrasting rotations are consistent with a collision of the Philippine arc with a northeastward-converging Palawan block.

INTRODUCTION

The Philippine archipelago has a complex history—with possible arc-arc collisions, flipping of subduction zones, and superposition of younger arcs over older inactive arcs—which makes it difficult to reconstruct the geologic history of the area. In the central Philippines, northern Palawan and the small Busuanga islands to the east (Fig. 1, inset) have a Paleozoic basement that is in fault contact with the southern half of Palawan. This older Paleozoic terrane is considered by Hamilton (1979) to be a continental fragment

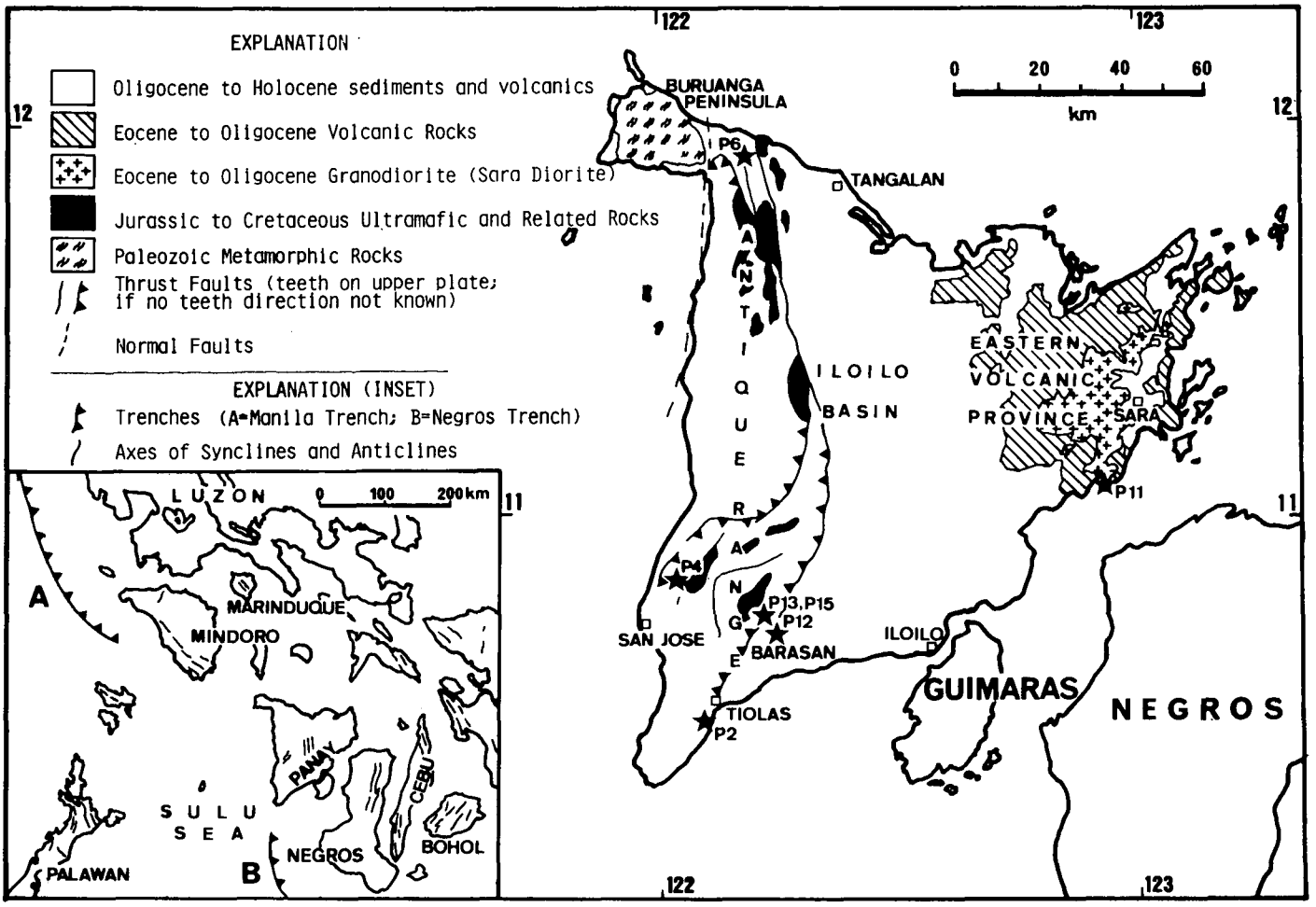


Figure 1. Simplified geologic map of Panay, showing location of four different tectonic provinces. Stars indicate paleomagnetic sites. Location map (inset) shows trends of axes of Miocene folds on islands of central Philippines. Data taken from Philippine Bureau of Mines (1964) and Balce (unpub. map). Note: Granitoid in Eastern Volcanic Province is tonalite.

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because of its anomalous location and the fact that it is composed of metamorphic rocks whose protoliths appear to be continental in origin.

The Manila Trench and the Negros Trench are zones of active subduction; however, in the central region, where this metamorphic terrane intersects the trace of these two zones of subduction, there is an area that appears not to be undergoing active subduction. The island of Panay, located in this central region, is characterized by a lack of volcanic activity. The seismicity around this island is dominated by shallow earthquakes that do not show thrusting focal mechanisms (Cardwell and others, 1980).

We suggest that western Panay is an area in which a late Oligocene to middle Miocene collision occurred between the older continental fragment and a Miocene west-facing subduction zone. This hypothesis is supported by both geologic and paleomagnetic evidence.

GEOLOGY OF PANAY

The island of Panay can be separated into four different tectonic provinces (Fig. 1):

1. The eastern province is composed of a basement of probable middle Tertiary arc-related volcanic and sedimentary rocks that are intruded by a possible upper Eocene to lower Oligocene (F. F. Florendo, personal commun.) hornblende tonalite, the Sara Diorite. These rocks are overlain by volcaniclastic rocks, volcanic flows, and sedimentary rocks that contain middle to late Miocene fossils (Capistrano and Magpantay, 1958).

2. The Iloilo basin, situated to the west of the eastern volcanic province, contains a thick sequence of Oligocene to Holocene sedimentary rocks. In the western part of the basin, the upper Oligocene–lower Miocene strata are composed of coraline limestones, coarse clastic sediments, and interbedded augite olivine basalt flows. Upsection, the middle Miocene to Pliocene strata change to sequences of fine-grained sandstone and shale. This younger section has no volcanic flows. Ohara (1969) analyzed the heavy minerals in the Iloilo basin sediments and found that the lower to middle Miocene sediments were derived from a volcanic source terrane, whereas the upper to middle Miocene sediments are depleted in both augite and hornblende, suggesting a waning of volcanic sources.

3. Reconnaissance mapping of western Panay (Jagolino and Jandumon, 1973; Diegor, 1980; R. McCabe, unpub.) shows that the Antique Range is composed of a polymict melange that contains blocks of glaucophane schist, gabbro, peridotite, metavolcanics, and red chert embedded in a serpentinite matrix. West of this melange belt is a series of north- to northwest-trending thrust faults. There, coherent blocks of Upper Jurassic–Lower Cretaceous (Matsuyama and McCabe, unpub.) red chert, peridotite, and metabasalt were thrust over lower to middle Miocene limestone and tuffaceous sediments. These units are overlain by undeformed fossil-bearing calcareous sediments of Pliocene age. The relationship of these sediments and the fault contact between the ophiolitic rocks and the Miocene sediments suggest that thrusting occurred between middle Miocene and Pliocene time.

4. A north-trending, probably fault-controlled valley at the neck of the Buruanga Peninsula separates the westernmost terrane from the Antique Range. Situated 50 km south of the peninsula are a few small islands with basement similar to the peninsula, indicating that this north-trending fault may extend along most of the western coastline of Panay. A brief examination of the basement on the peninsula shows that it is composed of schistose rocks and marble, which are intruded by a biotite-quartz diorite pluton. Although the age of this pluton is unknown, its foliation suggests that it may be similar in age to the rest of the highly foliated basement. A cursory petrographic examination of rocks from this terrane shows that they are siliceous schist and marble schist whose protoliths were fine quartz-rich sandstones and limestones. This supports Hamilton's (1979) conclusion that this block is continent-derived and related to the Palawan block.

TABLE 1. BEDDING-CORRECTED PALEOMAGNETIC RESULTS FROM PANAY

Site	Estimated age (Gonzales, 1963)	Rock type	Bedding Strike	Bedding Dip	N	Demagnetization level AFD(Oe)	Demagnetization level ThD(°C)	Dm	Im	α_{95}	k
P-11	Late Eocene	Tonalite	—	—	8	150	**	72.7	20.0	11.7	23.5
P-13	Late Oligocene-Early Miocene	Calcareous Sediment	N45°E	52°SE	11	200	450	-164.5	-16.6	5.7	65.0
P-15	Late Oligocene-Early Miocene	Basalt	N45°E	52°SE	15	200	450	-157.8	-18.7	4.2	82.7
P- 2	Middle Miocene	Siltstone	N16°W	45°SW	7	100	350	10.0	24.3	16.6	12.1
P- 4	Middle Miocene	Tuffaceous Siltstone	N27°E	38°SE	10	200	350	34.0	23.9	6.7	23.9
P-12	Middle Miocene	Calcareous Sediment	N45°E	62°SE	9	100	450	26.3	26.1	17.2	9.9
P- 6	Middle Miocene	Shale	N40°W	13°SW	6	100	—	10.3	19.9	12.0	32.2
Mean value for sampling sites (except P-11)								20.9	22.1	8.0	70.3

N, number of samples; AFD and ThD, alternating field and thermal demagnetization; Dm and Im, mean declination and inclination; α_{95}, radius of 95% confidence circle; k, precision parameter; **, not stable to ThD. Site locations: P-11, at the end of the road on a small peninsula located on the southeast part of Panay; P-13 and P-15, 6 km northwest from the Barrio Barasan along a small river; P-2, 60 km from Iloilo along highway 1; P-4, a road cut 25 km northeast from San Jose; P-12, 5 km down stream from P-15; P-6, 6 km west of the town of Tangalan along highway 1.

PALEOMAGNETISM

Hsu (1972) carried out the first paleomagnetic work in the Philippines. His results showed that during Tertiary time the islands of Luzon and Marinduque rotated counterclockwise and migrated northward. However, because of poor age control, Hsu was unable to discern the timing of the rotation. Later workers (Dunn and others, 1979; McCabe and others, 1980; Fuller and others, 1982) showed that the Oligocene to middle Miocene magnetic directions are distinguishable from the present field, being rotated between 20° and 30° counterclockwise with no significant difference in inclination values from the present field. Fuller and others (1982) have shown that the declination values for late Miocene to Pleistocene sites from Marinduque and Luzon are indistinguishable from the present field, suggesting that the counterclockwise rotation of the northern Philippines was completed by the late Miocene.

To augment the above studies, samples were collected from fifteen sites in Panay (Fig. 1). Five to six oriented hand samples were collected from each site and 25-mm cores were drilled from each sample and measured with a spinner magnetometer (Schonstedt SSM-1A). Three to four cores from each site were stepwise demagnetized by alternating field (AF) demagnetization. The step that gave the least scatter was chosen as the optimum value for the demagnetization of the remaining cores. Pilot samples were also demagnetized by stepwise thermal demagnetization (at site P-6 only six cores were obtained; all of these cores were treated by AF demagnetization only). From the paleomagnetic study, seven sites gave results that were consistent within the individual site; only these seven sites will be discussed further. The overall results of these measurements are represented in Table 1.

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Tilt-corrected directions, determined for the individual samples from each site, are plotted on equal-area projections (Fig. 2); a pair of projections are presented for each site. The magnetically cleaned plots show a strong convergence of the magnetic directions. This convergence indicates that the cleaning techniques are effective in removing the randomly directed secondary components of magnetization. This is also suggested by the presence of both normal and reversed polarity sites whose directions are antipodal at the 95% confidence level. These facts (clustering after magnetic cleaning and the presence of normal and reversed sites) plus the fact that the sites give consistent directions to both AF and thermal demagnetization support the conclusion that the directions reported are the primary directions of magnetization.

The mean direction of magnetization for the late Oligocene to middle Miocene sites from Panay (tilt corrected) is declination 20° E and inclination 22° down. When compared to the present dipole direction for Panay, a discordance in declination of 20° clockwise is noted. When the declinations of Panay rocks are compared with the results from Luzon, a discordance of between 40° and 50° clockwise is observed.

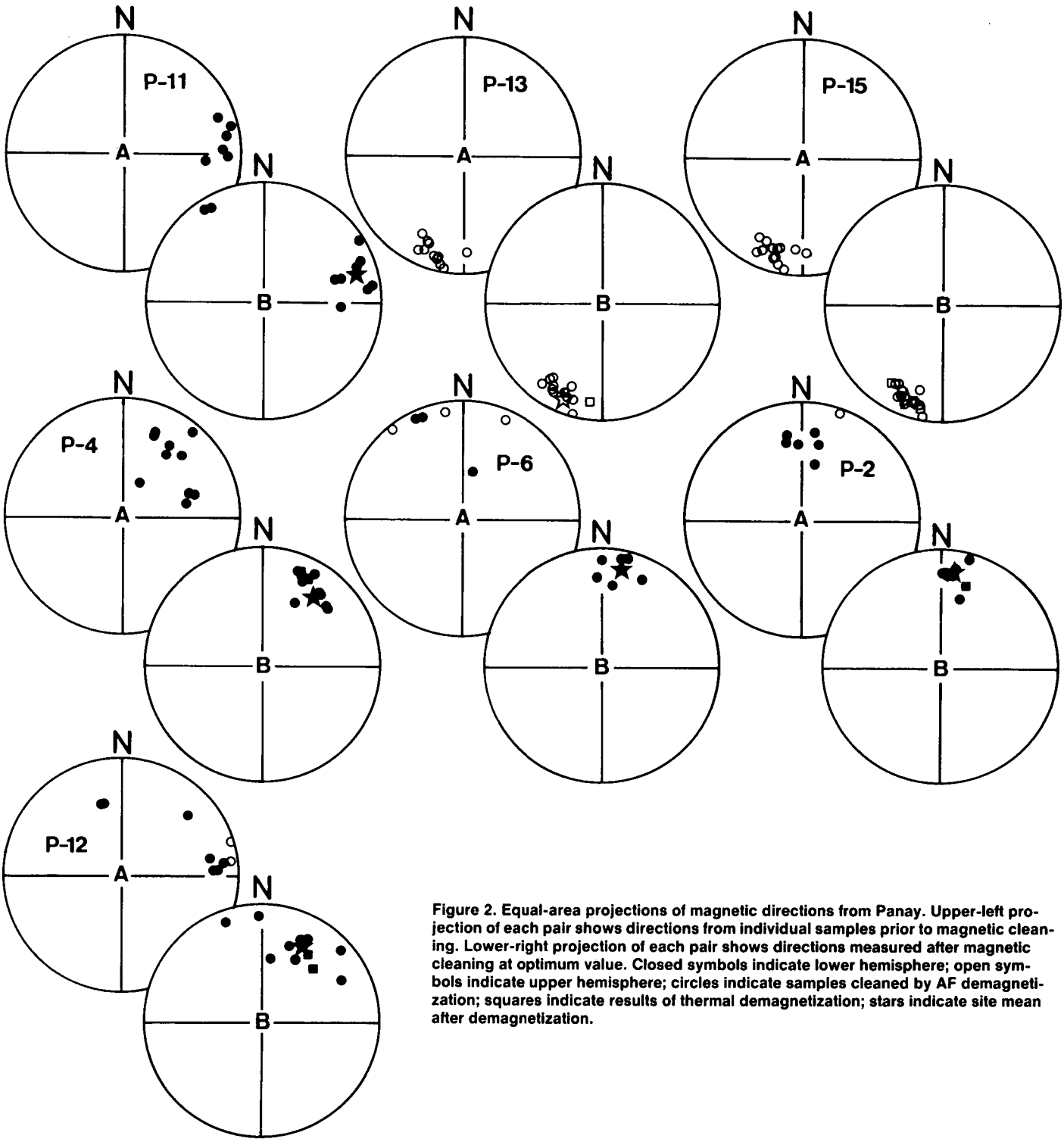


Figure 2. Equal-area projections of magnetic directions from Panay. Upper-left projection of each pair shows directions from individual samples prior to magnetic cleaning. Lower-right projection of each pair shows directions measured after magnetic cleaning at optimum value. Closed symbols indicate lower hemisphere; open symbols indicate upper hemisphere; circles indicate samples cleaned by AF demagnetization; squares indicate results of thermal demagnetization; stars indicate site mean after demagnetization.

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Although there are many possible explanations for this or any set of discordant paleomagnetic results, we favor the speculative model below; it is consistent with both the paleomagnetic and geologic evidence from Panay and the major structural trends of the central Philippines.

TECTONIC DEVELOPMENT AROUND A COLLISION ZONE

A hypothetical schematic time sequence is illustrated in Figure 3. At time T_1, subduction of an oceanic plate A is in progress at trench-island arc complex I. Located on plate A is a nonsubductable low-density fragment B (such as a piece of rifted continent). In this model we assume that plate A is stationary. Subduction of A at I eventually brings I in contact with B at T_2. As a result of the collision, horizontal forces directed normal to the line of intersection of fragment B and trench I (hereafter called the collision zone) will increase. During the same period, the down-directed vertical forces responsible for subduction are opposed by the buoyancy of fragment B (Vogt and others, 1976). These changes in the stress regime along the plate boundary may lead to the following tectonic events in the vicinity of the collision zone: (1) subduction will be locally halted (Vogt and others, 1976), (2) subduction-related volcanism around the collision zone will be inhibited (Nur and Ben Avraham, 1982), and (3) the horizontal compression, caused by the collision, will be transmitted across the plate boundary. This will create a broad zone of deformation, including the development of fold and thrust belts (Molnar and Tapponier, 1975; Matsuda, 1978).

With continued subduction of plate A, the active trench, to the north and south of the collision zone, will continue to move west relative to B (T_3). The margins of plate C, adjacent to the northern and southern edges of B, will rotate counterclockwise and clockwise, respectively. As this system continues to evolve, stresses will begin to accumulate between the actively subducting parts of the trench and the inactive part of the trench pinned in the collision zone. These increased stresses will eventually cause the development of either strike-slip faults along the east-west–trending boundaries of B (Vogt and others, 1976) or reorganization of the entire plate boundary.

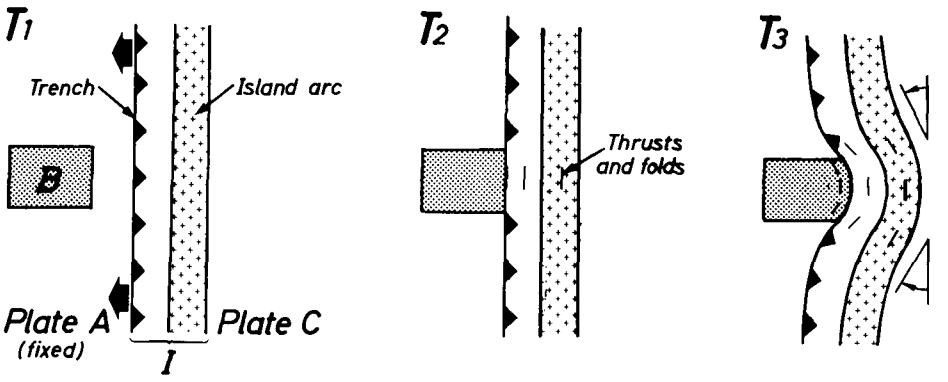


Figure 3. Schematic time sequence of island arc–trench complex I (located on the western margin of plate C), approaching nonsubductable continental fragment B on oceanic plate A. Plate A is assumed stationary. At T_2, trench I collides with B. After collision, B locally arrests westward migration of plate C. This results in compressional forces that are directed normal to line of intersection of B and plate C (collision zone). This compression results in development of fold and thrust belts on both fragment B and adjacent area of plate C. Segments of I north and south of collision zone will continue to migrate to left. This migration will result in tectonic rotations shown at T_3.

DISCUSSION

Although numerous complications exist, the model above is consistent with many of the large-scale tectonic features observed in the central Philippines.

In western Panay, the Buruanga Peninsula metamorphic terrane is juxtaposed against a high-pressure melange basement of the Antique Range. This peninsula, together with parts of Mindoro and the northern section of Palawan, constitutes part of a continental fragment (Palawan block). The Permian to Paleogene rocks of the Palawan block indicate that the geologic origin of this block is unrelated to the remainder of the Philippines (Hamilton, 1979; Holloway, 1982).

In the Antique Range, Mesozoic basement is thrust over lower to middle Miocene strata along north-trending reversed faults. Stratigraphically above these thrust rocks are undeformed Pliocene corals. These observations indicate that during the Miocene, the west coast of Panay was subjected to strong east-west compression. Other evidence for east-west compression is observed throughout the central Philippines: (1) on Mindoro and Marinduque, Miocene strata are folded with axes trending northwest (Fig. 1, inset); (2) on both of these islands, Mesozoic or lower Tertiary strata are thrust over Miocene on northwest-trending thrust faults; (3) on islands located south of Panay, (Cebu, Negros, and Bohol), Miocene strata are folded with axes trending north-northeast; (4) on Cebu, Cretaceous rocks are thrust over Miocene sediments along north-northeast–trending reversed faults (G. R. Balce, unpub. map). The Pliocene rocks on all islands discussed above are essentially nondeformed (Balce and others, 1981). All of these observations suggest that sometime during the Miocene the entire west-central Philippines were subjected to east-west compression.

Paleomagnetic data are also consistent with the model above. Although at present no stable reference frame exists against which to compare the magnetic directions of the Philippines, the results show that Luzon has rotated 40° to 50° counterclockwise relative to Panay since the Oligocene. Late Miocene to Pleistocene sites from Luzon and Marinduque, in the northern Philippines, and Negros, in the southern Philippines (McCabe, unpub. data), show no rotation relative to the dipole field during the past 10 m.y. These results suggest that between late Oligocene and late Miocene time the northern Philippines rotated counterclockwise relative to the central Philippines. These rotations are consistent with the proposed collision.

The above paleomagnetic and geologic evidence supports the following scenario for the Oligocene to Holocene evolution of the central and northern Philippines (Fig. 4). Oligocene to Miocene felsic plutonic rocks form a nearly continuous belt from northwestern Luzon (Balce and others, 1981) to Negros (A. Momongan, personal communs.). This belt of rocks suggests that during the Oligocene, active subduction was occurring along the western edge of the Philippine archipelago. We suggest that sometime during the late Oligocene to middle Miocene, the Palawan block, situated on the downgoing plate, collided with the central Philippines. This collision is supported by (1) ophiolitic rocks thrust over Miocene sedimentary rocks in western Panay; (2) middle Miocene waning of volcanic activity in eastern Panay (Ohara 1969); (3) juxtaposition of the Buruanga Peninsula against the Antique Range high-pressure metamorphic terrane; and (4) the

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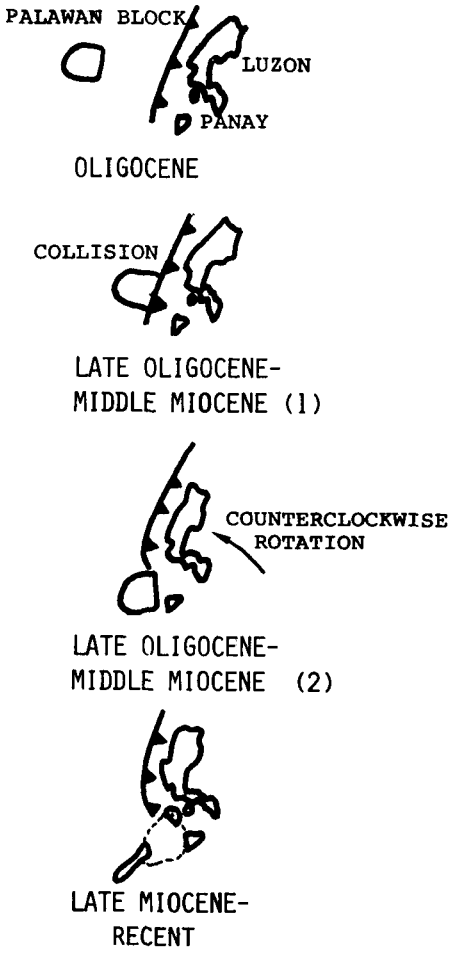


Figure 4. Oligocene to Holocene tectonic history of central Philippines. Oligocene—East-dipping subduction zone existed off west coast of Luzon and central Philippines. Palawan block is west of subduction zone. For simplicity, southward motion of Palawan block and opening of South China Sea (Taylor and Hayes, 1980) are not shown. Late Oligocene to middle Miocene—(1) Collision of Palawan block with central Philippines occurs. As result of this collision, fold and thrust belts develop in central Philippines (see Fig. 1 inset for orientation of Miocene fold belts). (2) At some undetermined later time (within this time span) northern Philippines is rotated counterclockwise relative to Panay. Late Miocene to Holocene—Paleomagnetic data from Luzon, Marinduque, and Negros suggest that northern and central Philippines have behaved as single tectonic block during this period. Dotted line shows location of accreted Palawan microcontinent.

occurrence of extensive Miocene folding throughout the central Philippines, with fold axes trending in directions that are consistent with the proposed collision (Fig. 1, inset). As a result of the collision, the northern Philippines rotated counterclockwise. This rotation is seen in the paleomagnetic results and in the trends of fold axes on Mindoro and Marinduque.

Late Miocene to Pleistocene paleomagnetic results suggest that the collision-related rotations of the Philippines were completed after the middle Miocene. The fact that Luzon, Marinduque, and Negros rocks show consistent late Miocene magnetic directions suggests that since the late Miocene the entire northern and central Philippines have behaved as a single tectonic block.

In the model discussed above, we have used Panay as our fixed reference point against which to compare the paleomagnetic directions of the northern Philippines. We are not suggesting that the clockwise rotation of Panay is not related to the collision. In fact, Panay's location (adjacent to the southern boundary of the Palawan block) suggests that Panay has undergone some rotation as a result of the collision. Further studies should be aimed at collecting more samples from Panay and the other islands in the southern archipelago.

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